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CAM-Former: A Cooperative-Attention-based Mamba Transformer for Multi-Source Remote Sensing Land Cover Classification

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Abstract—To address the insufficient global modeling capability of the Mamba model in multi-source remote sensing data joint classification, as well as the challenges of modality discrepancies and uneven weight assignment in multi-source data fusion. This paper presents the Cooperative-Attention-based Mamba Transformer(CAM-Former). Firstly, the Mamba-based Transformer for Cross-Modality Integration (MambaFormer-X) is designed to augment the model's global feature extraction capability by leveraging Transformer's self-attention mechanism, while preserving Mamba's strengths in modeling long-range dependencies. Secondly, this paper introduces the Cooperative Attention Feature Aggregation (CAFA) module to address key challenges in multi-source data fusion, including inter-modal discrepancies, imbalanced spatial significance, and non-adaptive weight allocation. Finally, experiments on two real-scenario datasets show that this network outperforms the current state-of-the-art networks.

Index Terms—multi-source; Mamba; Transformer; Cooperative Attention.

I. Introduction

The utility of Hyperspectral Imaging (HSI) and Light Detection and Ranging (LiDAR) or Synthetic Aperture Radar (SAR) data analysis for feature classification is well established in numerous domains, including the monitoring of land cover and use, the detection of surface changes, and geological research [1], [2].

Early studies in the field largely depended on single-sensor data, especially HSI. However, data from a single sensor are often limited in aspects such as spatial and spectral resolution. Multi-source remote sensing data fusion has thus been developed to integrate information from different sensors, leading to more accurate land cover classification [3]. LiDAR delivers high spatial resolution and accurate 3D structural measurements. SAR demonstrates exceptional all-weather imaging capability, maintaining consistent image quality unaffected by external conditions such as cloud cover or sandstorms, thereby enabling reliable acquisition of high-resolution terrestrial imagery. The fusion of HSI, and LiDAR over common geographic coverage—can improve land cover classification accuracy. That said, joint classification of HSI and LiDAR data remains a challenging research area.

Deep learning-based joint classification frameworks typically involve three key stages: feature extraction, multi-source feature fusion, and final classification. A number of deep learning models have been proposed for joint classification, with convolutional neural network (CNN) architectures being among the most widely adopted. Effective fusion of HSI and LiDAR requires careful consideration of their differing characteristics. Experiments [3] have shown that dual-branch CNN designs can significantly improve model performance.

While CNNs are well-suited for spatial feature extraction, they are less capable of handling the long spectral sequences in HSI. The transformer have recently exhibited strong performance in multi-source remote sensing classification, due to their ability to global dependency modeling capability, leverage hierarchical attention, and integrate spatial-spectral information through self-attention mechanisms [4]–[6]. They benefit from a global receptive field and are effective at capturing long-range relationships. However, they are computationally inefficient when processing long sequences and involve high-complexity attention operations.

Recently, Mamba has been introduced as a new deep learning architecture, which achieves a balance between computational efficiency and global context integration [7]. By generalizing the original unidirectional scan to

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multi-directional scanning, Mamba-based models have exhibited better performance in image processing tasks [8], [9]. With an expanded receptive field, Mamba can capture wider spatial dependencies, leading to more effective receptive field modeling compared to both CNN and spatial-spectral Transformer approaches. While Mamba is capable of establishing long-range dependencies, its global modeling capability is limited.

II. The Method

A. The Overall Framework

As shown in Figure 1, this paper employs two subnetworks to process HSI and LiDAR / SAR obtained from the same geographical area in parallel. The HSI branch (HSI-Net) processes the HSI data with dimensions $X_H \in \mathbb{R}^{h\times w\times c}$, while the LiDAR branch (LiDAR-Net) processes the LiDAR data with dimensions $X_L \in \mathbb{R}^{h\times w}$.

During the preprocessing stage, the Principal Component Analysis (PCA) is applied to decrease the dimensionality of HSI, firstly. The dimensionality reduction process strikes an effective balance between maximizing the retention of critical information in the HSI data and the elimination of redundancies, thus providing an optimal basis for further processing. The value of c was set to 30 based on experimental results.

$$X_{\mathrm{H}} = PCA(HSI) \tag{1}$$

Subsequently, both the LiDAR and the dimensionality-reduced HSI are divided into image patches of size p×p, which serve as the input to the network.

The HSI-Net employs a hybrid 3D-2D convolutional structure: a 3D Convolutional Neural Network (3D-CNN) is utilized to extract joint spectral-spatial features from HSI, while a 2D Convolutional Neural Network (2D-CNN) is adopted to capture local texture information and establish dependencies among multi-channel features. Following each convolutional layer, a Leaky ReLU activation function and Batch Normalization (BN) are incorporated to enhance gradient stability and improve model convergence efficiency. For the LiDAR-Net, spatially co-registered patches corresponding to HSI are directly extracted. To mitigate potential overfitting, batch normalization (BN) is synchronously applied throughout this network training process.

B. The MambaFormer-X

To address the limitations of the Mamba in global modeling capacity and attention mechanism performance, this paper proposes the Mamba-based Transformer for Cross-Modality Integration (MambaFormer-X).

During the processing of HSI, the input HSI feature map is shown as follows:

$$X_{\mathrm{HSI}} \in \mathbb{R}^{B\times C\times L} \tag{2}$$

where B represents the batch size, C denotes the number of channels , and L corresponds to the sequence length.

It is first modeled using a multi-head self-attention (MHSA) mechanism to capture spectral dependencies. Specifically, we compute the Query, Key, and Value representations via linear transformations:

$$Q = W_Q X_{\mathrm{HSI}}, \quad K = W_K X_{\mathrm{HSI}}, \quad V = W_V X_{\mathrm{HSI}} \tag{3}$$

Then, the self-attention mechanism computes the weighted feature representation:

$$\mathrm{Attn}(Q,K,V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V \tag{4}$$

Finally, different spectral patterns are aggregated through the multi-head mechanism:

$$H_{\mathrm{HSI}} = W_O[\mathrm{Attn}_1, \mathrm{Attn}_2, \ldots, \mathrm{Attn}_h] \tag{5}$$

where h is 8. This process effectively models the spectral characteristics of HSI while mitigating the limitations of conventional convolutional approaches, which may overlook critical spectral features. Consequently, the model gains a deeper understanding of the internal correlations within HSI.

In contrast to HSI, which captures extensive spectral information, LiDAR predominantly conveys elevation information, often exhibiting a higher signal-to-noise ratio and more stable local features. Consequently, its feature extraction can be effectively achieved through Convolutional Neural Networks, circumventing the necessity for supplementary Transformer-based self-attention mechanisms.

For LiDAR, the input LiDAR feature map is defined as:

$$X_{\mathrm{LiDAR}} \in \mathbb{R}^{B\times C\times L} \tag{6}$$

where B represents the batch size, C denotes the number of channels, and L corresponds to the sequence length. Since it has already undergone feature extraction via a convolutional neural network, additional Transformer-based processing is unnecessary. Instead, the extracted features are directly fed into the Cross-Modality Mamba module:

$$H_{\mathrm{LiDAR}} = X_{\mathrm{LiDAR}} \tag{7}$$

This design choice aims to preserve the spatial structural information of LiDAR while avoiding unnecessary self-attention computations, thereby reducing computational complexity and improving inference speed. Since LiDAR primarily provides spatial structural information rather than spectral characteristics, there is no need for spectral modeling via a Transformer. Instead, Mamba is employed for efficient feature fusion. Ultimately, both are collectively input into the Synergistic-Mamba.

By incorporating the self-attention mechanism of the Transformer, the MambaFormer-X's ability to capture

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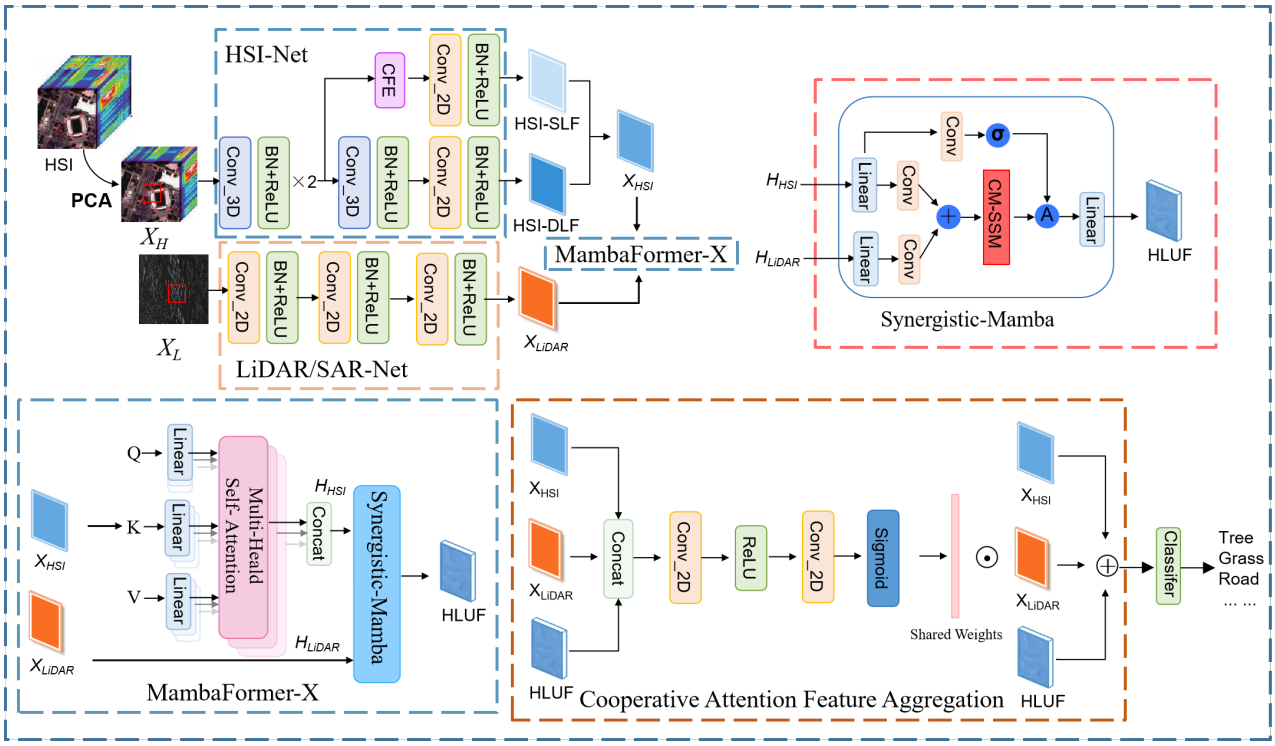


Fig. 1. The Cooperative-Attention-based Mamba Transformer(CAM-Former)

global features is enhanced while preserving Mamba's advantage in modeling long-range contextual dependencies. Specifically, The Transformer's self-attention mechanism can capture global contextual information within the input data, including dependencies across spatial and channel dimensions, further improving the capacity of the network for multi-modal data representation. The self-attention mechanism explicitly models the spatial structural information present in HSI and LiDAR, strengthening its ability to characterize spatial dependencies. The cross-modal state space model in Cross-Modality Mamba continues to leverage its strength in efficiently modeling long-range dependencies.

C. The Cooperative Attention Feature Aggregation (CAFA)

To address the challenges of multi-source data fusion, including modality discrepancies, uneven spatial significance, and the difficulty of adaptively assigning weights, this paper introduces the Cooperative Attention Feature Aggregation (CAFA) module.

In the processing stage, the three inputs are first concatenated along the channel dimension to form a composite feature tensor. This tensor is then passed through a lightweight attention-generating network composed of two 1×1 linear convolutional layers, with a ReLU activation inserted between them to enhance nonlinearity. A final Sigmoid function produces a spatial attention map with values in the range [0,1], indicating the relative importance of different spatial locations.

During feature fusion, the generated attention map is applied element-wise to each of the three inputs, thereby assigning adaptive weights to different modalities. The weighted results are subsequently aggregated by averaging, producing the final fused representation. This design allows the network to modulate the contribution of each modality in accordance with the input characteristics, while ensuring consistency across data sources and emphasizing informative features.

III. The Experiment

1) Experiments Setting: All experiments in this paper were carried out on a single computing platform. The system was equipped with an NVIDIA RTX 4090 graphics processor with 24 GB of memory, a 16-core central processing unit, and 120 GB of system memory. The software environment included CUDA 11.8, and Python 3.8. During the training process, the learning rate was set to 0.0001, the batch size was 128, and the procedure was repeated for 100 epochs.

2) Datasets: Houston 2013 [10]: The Houston 2013 dataset, collected by NCALM at the University of Houston in 2012, integrates LiDAR and HSI data with a spatial resolution of 2.5 meters and an image size of 349 × 1905 pixels.

Berlin [11]: The Berlin dataset comprehensively covers both urban and rural areas of Berlin, Germany, integrating multi-source remote sensing data from SAR and HSI acquired over the same geographic region. The dataset features a spatial dimension of 797 × 220 pixels.

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3) Evaluation Metrics: To quantitatively evaluate the classification performance of the proposed framework and compare it with existing models, this paper employs three widely recognized metrics: Overall Accuracy (OA), quantifying the proportion of correctly classified samples across all categories; Average Accuracy (AA), which evaluates the mean accuracy across all classes to reflect performance consistency; and Kappa Coefficient (Kappa), assessing the agreement between predicted and actual classifications.

4) Comparative Experiments on the Houston 2013 Dataset: A carefully designed series of experiments was carried out to rigorously evaluate the network's performance. [12]–[16].

TABLE I
Comparative experiments (%) on the Houston 2013 dataset.

Class	HybridSN	MDL	S^2ENet	EndNet	FustAt	Ours
C0	93.76	81.39	83.95	84.35	82.49	99.77
C1	87.41	99.66	100.00	96.99	99.91	95.98
C2	100.00	82.81	99.72	99.80	81.03	100.00
C3	98.35	99.21	99.73	97.57	97.44	96.86
C4	98.78	98.96	100.00	99.43	98.58	98.04
C5	99.00	98.46	98.60	95.66	94.82	99.30
C6	86.33	92.65	91.12	81.59	92.56	90.76
C7	88.33	89.82	91.41	78.29	81.35	94.16
C8	81.29	89.65	88.25	78.75	86.61	98.73
C9	81.32	78.51	92.22	80.17	78.65	73.66
C10	72.97	95.20	97.33	97.57	92.85	98.48
C11	92.25	75.81	87.69	87.92	88.26	96.66
C12	90.30	92.49	91.51	79.44	87.37	100.00
C13	98.37	98.79	98.38	100.00	89.88	100.00
C14	97.62	96.45	99.96	98.44	77.76	99.78
OA	88.50	90.40	93.84	89.26	89.06	94.44
AA	91.07	91.32	94.66	90.40	88.64	96.14
Ka	87.52	89.59	93.32	88.35	88.13	93.96

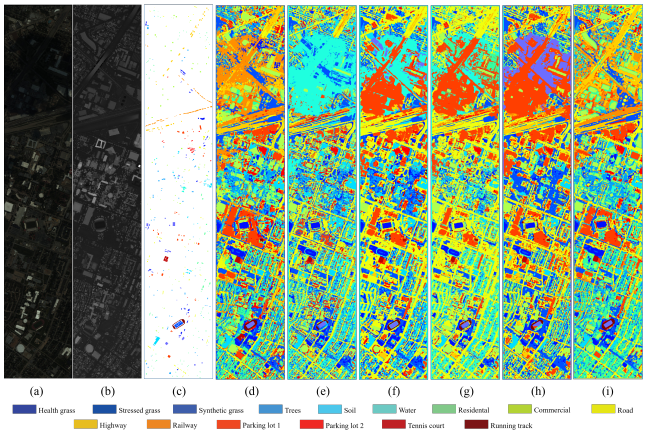


Fig. 2. The Classification maps by different methods on the Houston 2013 dataset. (a) HSI, (b) LiDAR, (c) Ground Truth, (d) HybridSN, (e) MDL, (f) S^2ENet, (g) EndNet, (h) FustAtNet and (i) Ours.

TABLE II
Comparative experiments (%) on the Berlin dataset.

Class	Hybrid	MDL	S^2ENet	EndNet	FustAt	Ours
C0	74.59	81.42	78.65	77.09	74.86	72.36
C1	66.69	74.66	84.07	71.05	79.64	81.26
C2	55.43	73.99	52.22	45.31	28.53	52.27
C3	64.76	83.24	75.67	62.15	47.95	72.34
C4	74.67	59.39	73.30	83.18	62.49	91.84
C5	62.69	67.32	48.58	70.89	55.09	46.93
C6	32.80	14.16	9.49	39.33	24.57	32.76
C7	59.17	48.86	71.97	79.98	53.13	58.64
OA	65.18	72.17	71.17	68.45	68.3	73.18
AA	61.35	62.88	66.28	66.12	53.28	85.32
Ka	50.91	59.59	58.55	55.27	51.5	87.08

As shown in Table 1 and Figure 2, CAM-Former achieves optimal or near-optimal classification accuracy across multiple land cover categories, with particularly strong performance in spectrally similar classes(C6-c10) such as residential areas, commercial zones, roads, highways, and railways. This performance advantage stems from the model's synergistic integration of Transformer's global attention mechanism and Mamba's long-range dependency modeling, which significantly enhances its capacity for global feature extraction from multi-source remote sensing data.

As shown in Table 2 and Figure 3,Compared to other networks, MambaFormer-HCMNet demonstrates outstanding performance on the Berlin dataset, particularly in terms of overall classification accuracy, class balance, and effectiveness in specific categories. First, as observed from the tabulated data, our method excels in OA, AA and Kappa. Notably, it achieves the highest OA (73.18%) and Kappa (60.04%), indicating that our network attains optimal accuracy and consistency in the classification task. Although the AA (63.55%) is not the highest, it remains highly competitive, highlighting the robustness and generalizability of our approach in multi-source remote sensing applications. This performance advantage stems from our network's effective integration of Mamba's exceptional long-range dependency modeling capacity with the global contextual representation capabilities of the Transformer architecture, leading to a substantial enhancement in overall recognition performance.

TABLE III
The Ablation Analysis of MambaFormer-X (%) on the Houston 2013 dataset.

	OA	AA	Kappa
Base	91.60	93.67	90.88
Base + MambaFomer	94.17	95.58	93.67
CAM-Former	94.44	96.14	93.96

5) Ablation Experiments: As shown in Table 3, ablation experiments on the Houston 2013 dataset confirm that incorporating the MambaFormer module yields improvements of 2.57% in OA, 1.91% in AA, and 2.79% in Kappa. These results substantiate the module's significant advantage in enhancing classification performance. By maintaining Transformer's global modeling capacity while effectively leveraging inter-modal complementarity and synergistic effects, MambaFormer substantially strengthens the model's overall capability.

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IV. Conclusion

This paper presents CAM-Former. The MambaFormer achieves enhanced global feature extraction from multi-source remote sensing data through synergistic integration of Transformer's global attention mechanism and Mamba's long-range dependency modeling capabilities. The Cooperative Attention Feature Aggregation module resolves spatial importance imbalances in multi-source data fusion via dynamic weight allocation.

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